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EDUCATION

University of Bucharest

October 2024 – May 2028

B.S.E. Computer Science

- **Coursework:** OOP, Data Structures and Algorithms, Databases, Operating Systems, Multivariable Calculus and Linear Algebra, Numerical Methods, Physics, Digital Electronics, Fundamentals of Computer Networks

EXPERIENCE

Programming Trainer

September 2022 – July 2023

Logiscool

Râmnicu Vâlcea, Romania

- Taught 3 groups of 5 teenagers each block-based coding, C programming, and hands-on LEGO robotics.

PROJECTS

Airline Management System

- Developed an airline management system in C++ applying OOP principles including inheritance hierarchies, design patterns, templates, and more. Full code and documentation available on GitHub.

Formula 1 Management Database

- **Architecture:** Developed a relational database for Teams, Drivers, Cars, Sponsors, GPs, Circuits, and Sessions with an optimized schema design.
- **Data Integrity:** Applied constraints to enforce rules and maintain consistent data. Created multi-table queries with GROUP BY/HAVING to generate performance statistics and seasonal standings.
- **Views:** Created multiple complex and updatable views with ON DELETE CASCADE to automate reports and maintain data consistency.
- **Interface:** Developed Oracle APEX CRUD interface for data management; planning migration to Java front-end for enhanced user experience.

System Monitor Web Server – Rust Workshop

- Developed a web-based system monitor using the Axum framework and Tokio runtime.
- Implemented endpoints for real-time health checks and system performance metrics.
- Used asynchronous programming in Rust to handle diagnostic data efficiently.

NXP Line Follower Hackathon – Honorable mention

- Assembled the autonomous robot from a specialized kit, performing manual soldering of the electronic components and mechanical construction.
- Used NXP software to configure robot navigation and test real-time tracking sensors.

Linux Command Simulation Projects

- Developed a C program simulating the Linux `diff` command, supporting recursive directory comparison (`-r`) and efficiently detecting missing or differing files.
- Developed a C program that simulates the behavior of the `ls` command in bash.

Smart Task Scheduler

- Designed and implemented a CLI-based scheduling program in C that creates optimized weekly plans based on task duration and deadlines using priority queues and greedy algorithms.

Heat Equation Solver

- Built a 1D and 2D heat equation solver in Python using finite differences. Constructed and solved linear systems for regular and L-shaped domains and verified correctness by analyzing numerical error and convergence.

Faculty of Mechanical Engineering and Mechatronics Hackathon – 3rd Place

- Built an obstacle-avoiding robot with Arduino and sensors, working in a team of 4 to design and test the navigation logic.

PageHaven – Full-Stack Virtual Library (Ongoing)

- **Architecture & Security:** Engineered a **JavaScript** backend using **Node.js** and **Express**, integrating **PostgreSQL** with secure user authentication, password hashing, session management, and role-based access control.
- **Frontend & UI:** Built a responsive interface utilizing **Bootstrap** and **EJS** server-side rendering, featuring full-stack form validations and an automated **SCSS-to-CSS** compilation pipeline.
- **Media & Optimization:** Integrated the **Sharp** library for automated responsive image processing (WebP generation) and implemented multilingual accessibility via **VTT video tracks**.

EXTRACURRICULAR ACTIVITIES

FIRST LEGO League

September 2022 – July 2023

- **Vice-champion** at the National Championship and qualified for **international** competitions in **Morocco**.
- Programmed LEGO robots in EV3-G and Python for autonomous missions, including obstacle navigation, mechanism activation, and object delivery.
- Led and presented the team's robot and Innovation Project "SUPERPOWERED" on renewable energy.
- Organized STEM events to promote these values in the country and encouraged young students to join this community.

SKILLS

Languages: C (intermediate), C++ (intermediate), Python (intermediate), SQL (beginner)

Soft Skills: Problem Solving, Self-learning, Critical Thinking, Passionate, Adaptability, Teamwork, Proactive

Certifications: IC3 (3 modules)